

Name _____ Period _____

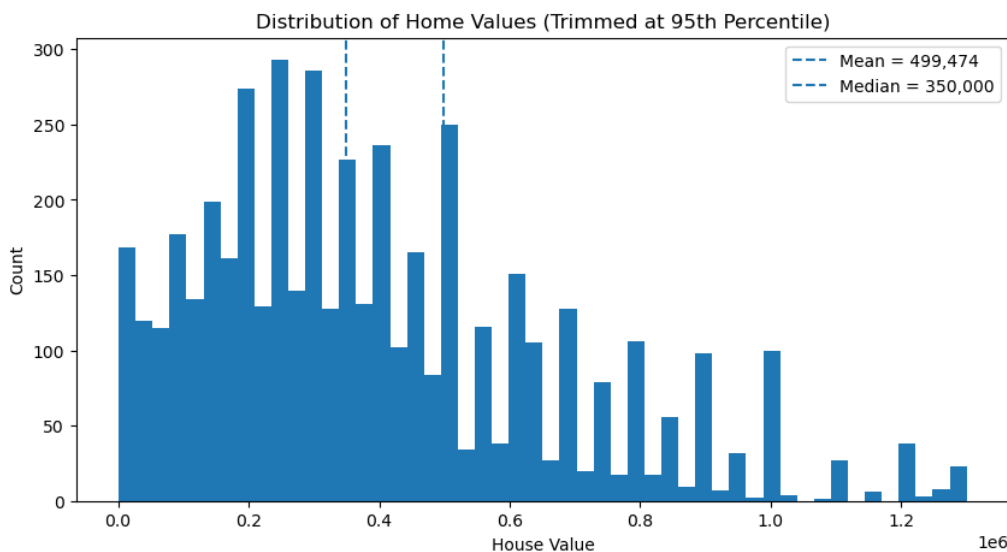
Skill 37.1 Exercise 1

For each situation, decide whether the mean or median is more appropriate and briefly explain why.

Situation	Mean/median	Explanation
The annual income of adults in New York	Median	
The heights of students at Timberline	Mean	
The selling price of homes in New York	Median	
The scores on a quiz in Pluska's class	Mean	
The number of Instagram followers	Median	

Skill 37.2 Exercise 1

A real estate company wants to describe a “typical” home value in a city. They collect data on home values and create the histogram below.



Based on the histogram, which statistic — the *mean* or the *median* — would be the more appropriate measure of center? Explain your reasoning

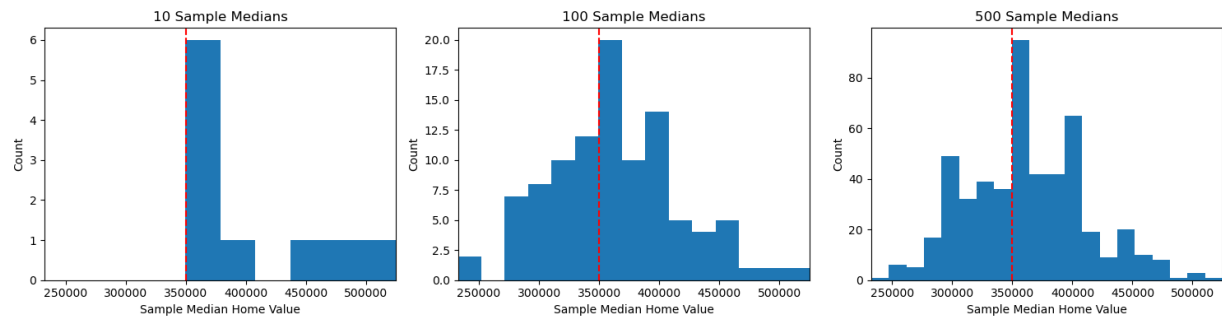
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Based on income data, city officials estimate that a typical home buyer in the city can afford a home worth \$350,000. They want to investigate whether a typical home in the city is more expensive than what a typical buyer can afford.

Write an appropriate null hypothesis and alternative hypothesis for this situation.

Skill 37.3 Exercise 1

A sample of homes has a median value above \$350,000. Based on the histograms below, explain why that result alone is not enough to conclude that the population median home value is greater than \$350,000.



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Skill 37.4 Exercise 1

Refer to the bootstrap function below,

```
import pandas as pd
import numpy as np

housing_data = pd.read_csv("housing.csv")

def bootstrap_median(original_sample, label, replications):
    just_one_column = original_sample[label]
    medians = []

    for i in range(replications):
        bootstrap_sample = np.random.choice(just_one_column,
                                             size=len(just_one_column),
                                             replace=True)

        resampled_median = np.median(bootstrap_sample)
        medians.append(resampled_median)

    return medians
```

Explain what the line highlighted in red is doing.

The code snippet below loads the housing data frame and displays the first few rows,

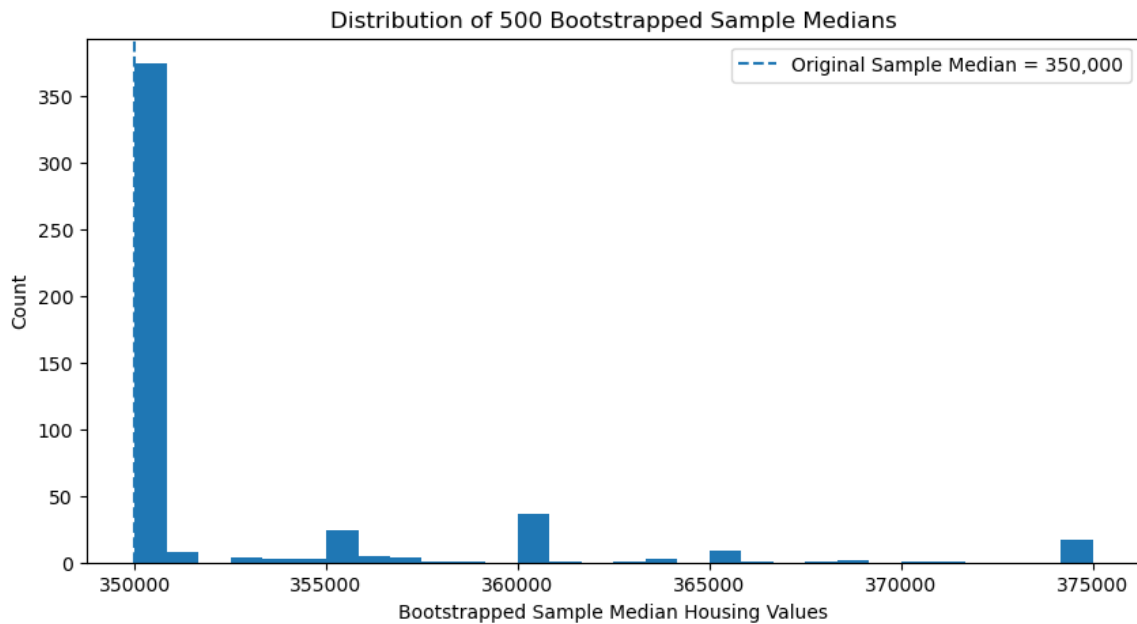
```
housing_data = pd.read_csv("housing.csv")
```

	YEAR	SAMPLE	OWNERSHP	VALUEH	HHWT
0	2024	202401	1	900000	62.0
1	2024	202401	1	300000	93.0
2	2024	202401	1	950000	41.0
3	2024	202401	1	300000	105.0
4	2024	202401	1	100000	13.0

Write a line of code that uses the function to create 500 bootstrapped sample medians for home values

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Below is a histogram of the samples



Why does the histogram have one very tall bar near \$350,000 and only a few bars farther to the right?

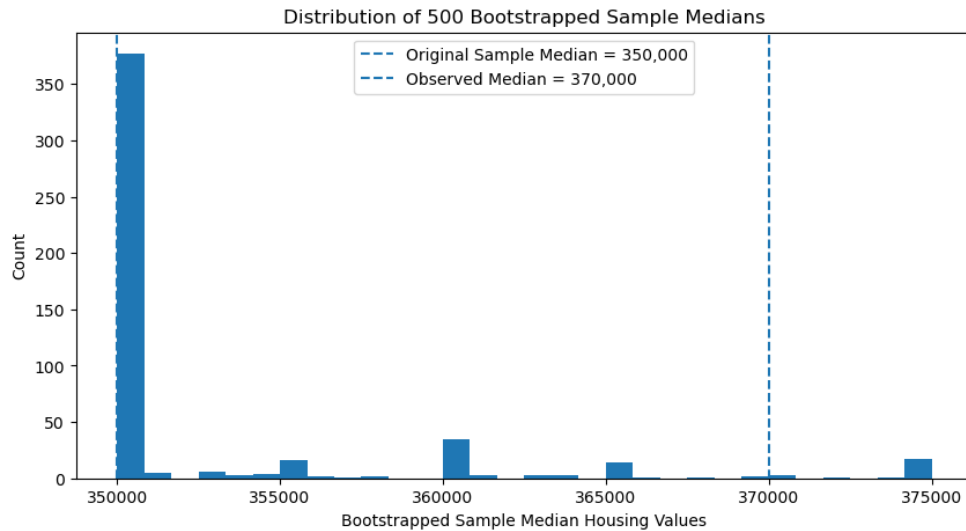
What does this tell us about the original housing dataset?

Suppose someone claimed that the population median home value was \$360,000. Would this histogram make that claim seem more believable or less believable? Explain

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Skill 37.5 Exercise 1

The histogram below shows a simulated distribution of 500 sample medians for home values under the null hypothesis that the population median is \$350,000. An observed sample median of \$370,000 is marked on the graph.



Looking at the histogram does a median of \$370,000 seem unusual?

The code segment below calculates the p value associated with an observed mean of \$370,000.

```
observed_median = 370000
count = sum(1 for median in boot_medians if median >= observed_median)
p_value = count / len(boot_medians)
print(p_value)
```

The p_value is 0.044

Write a sentence interpreting the p-value in context.

Is an observed sample median of \$370,000 common or unusual under the null hypothesis?

If the null hypothesis is that the population median home value is \$350,000, what conclusion would you make?

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